

# MANIKANDAA S

AI/ML Engineer | Python • TensorFlow • Scikit-learn

Tiruchirappalli, Tamil Nadu, India | +91-9445219410 | manikandaa944@gmail.com | [LinkedIn](#) | [GitHub](#)

## SUMMARY

Data Science graduate specializing in building machine learning systems end-to-end, from data pipelines and model training to integration into working applications. Hands-on experience with classical ML (Scikit-learn) and deep learning (TensorFlow/Keras), achieving up to 87% accuracy on healthcare prediction tasks. Comfortable writing modular, well-structured Python code and organizing ML workflows for maintainability. Strong statistical and data-engineering foundation, built through an M.Sc. in Data Science and hands-on internship experience with real-world government datasets.

## SKILLS

**Programming:** Python, R, SQL (MySQL)

**ML/DL Frameworks:** Scikit-learn, TensorFlow, Keras

**ML Engineering:** Model Training Pipelines, Feature Engineering, Hyperparameter Tuning, Cross-Validation, Modular Code Design

**Model Evaluation:** Accuracy, ROC-AUC, Confusion Matrix, Precision, Recall, F1-Score

**Data Engineering:** ETL Pipelines, Data Cleaning, Pandas, NumPy

**Tools:** Git, GitHub, Jupyter Notebook, Google Colab, VS Code

## EXPERIENCE

**Data Science Intern — Analytics Career Connect**

Apr 2024 – May 2026

Remote

- Performed data cleaning, preprocessing, and exploratory data analysis (EDA) on real-world datasets using Python, Pandas, and NumPy.
- Created data visualizations using Matplotlib to identify trends, patterns, and actionable insights.
- Applied statistical analysis and basic machine learning techniques to analyze datasets and uncover meaningful patterns.
- Developed analytical insights and communicated findings clearly through visualizations and concise reports.
- Prepared analytical reports translating technical findings into stakeholder-ready summaries to support data-driven decision-making.

## PROJECTS

**AI Job Recommendation System**

2025

Python · NLP · Scikit-learn · Application Development

- Engineered an end-to-end ML application matching candidate skill profiles to job roles, combining NLP-based text feature extraction with a trained classification model and cosine similarity matching.
- Structured the system into modular components (data processing, feature extraction, matching/inference), following clean software engineering practices for maintainability.
- Iterated on the matching logic using precision/recall evaluation to improve real-world relevance.

**Healthcare Prediction Model — Diabetes Detection**

2024

Python · Scikit-learn · Logistic Regression · Random Forest

- Built a complete supervised learning pipeline (preprocessing → EDA → feature selection → training → tuning → evaluation) on the PIMA Indians Diabetes Dataset.
- Trained and compared Logistic Regression and Random Forest models, achieving 87% accuracy with Random Forest after hyperparameter tuning.
- Validated generalization using k-fold cross-validation, ROC-AUC, and confusion matrix analysis.

**Cat vs Dog Image Classifier — Deep Learning (CNN)**

2024

Python · TensorFlow · Keras · Computer Vision

- Designed and trained a Convolutional Neural Network with convolution, max-pooling, and dropout layers using TensorFlow/Keras.
- Applied data augmentation (flip, rotation, zoom) and image preprocessing to improve model generalization on unseen inputs.

**Server Monitoring & Performance Data Pipeline**

2025

Python · ETL · Pandas · Power BI

- Engineered a modular ETL pipeline to ingest, clean, and transform server metrics data into structured formats for downstream analysis.
- Applied clean, single-responsibility code design across ingestion, transformation, and visualization layers — engineering practices directly transferable to production ML pipelines.

## EDUCATION

**Bharathidasan University — M.Sc. Data Science**

2024 – 2026

Tiruchirappalli, Tamil Nadu — Relevant coursework: Machine Learning, Deep Learning, Big Data Analytics, Data Mining, NLP, Statistical Computing

**St. Joseph's College — B.Sc. Statistics**

2021 – 2024

Tiruchirappalli, Tamil Nadu | Grade: 77.32% — Relevant coursework: Probability, Statistical Inference, Operations Research

## CERTIFICATIONS

Data Analytics using Python | Power BI – SkillNation | SPSS Certification & Data Mining Workshop | Fundamentals of Python